Online Examination System With Cheating Detection

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Chapter 1

Introduction:

Educational institutions at present encounter difficulties with their examination procedures which require them to secure their assessment methods and provide unbiased evaluation methods. The need for in-person supervision during examinations becomes impossible when students operate from different locations, resulting in increased cheating and making testing procedures difficult to manage.
The Online Examination System with Cheating Detection or (TESTify) will directly address this challenge by offering an integrated platform that enables institutions to create, deliver, and evaluate assessments all while incorporating automated proctoring and robust anti-cheating mechanisms to uphold academic integrity. [1]

Problem Statement

The rising popularity of distance learning has revealed fundamental weaknesses in examination security. Students can use unsupervised testing situations to find answers through online research and engage in forbidden teamwork and use illegal study resources. The existing manual testing methods lack both reliability and the ability to handle large-scale operations. The education system requires an advanced online examination system which uses artificial intelligence to detect and prevent cheating while maintaining a seamless experience for honest students.

Project Scope

The Online Exam System with Cheating Detection (OESCD) shows how its web-based platform provides secure online testing environments which protect students, instructors, and administrators. The system includes student registration and authentication, exam creation and scheduling, question bank management, AI-based proctoring (facial recognition, eye tracking, tab switch detection), browser lockdown enforcement, webcam monitoring, suspicious behavior logging, automated grading, result publication, and admin reporting. The system operates with dedicated lockdown extension which supports Chrome and Edge browsers. The system excludes three functions which involve physical examination management, student LMS integration, and payment gateway for exam fees. [2]

Functional Requirements

User Management

- The system shall allow new students and instructors to register with valid institutional email addresses.
- The system shall authenticate users using username and password
- The system shall allow admins to create, edit, deactivate, and delete user accounts.
- The system shall support role-based access control with roles:
  Student, Instructor, Examiner, Admin.
- The system shall allow users to reset their passwords via email verification.

Exam Management

- The system will enable different types of questions such as multiple-choice, true/false, short answer, and essay.
- The system will have a bank of questions that will be reused according to their subjects, topics, and difficulty level.
- The system will allow random selection (when required) of questions from the bank of questions.
- The system will automatically set time limits for the examination.
- The system will allow students to move forward and backward between questions and flag questions.
- The system will show the countdown timer during the entire examination period.

Cheating Detection

- The system must identify the student's face using the webcam prior to and during the exam.
- The system must identify eye movement that could be indicative of off-screen references being used.
- The system must identify tab switching and attempts to minimize windows and share screens.
- The system must force the browser into full screen mode and disallow other access.
- The system must identify multiple faces identified by the webcam and classify as cheating.
- The system must automatically produce a cheating incident report for each student and exam.
- The system must allow the instructor to view the cheating logs and recordings after the exam.

Grading & Results

- The system should be able to automatically evaluate multiple choice and true-false type questions.
- The system should enable teachers to manually assess short answer and essay type questions.
- The system should compute and display final grades with individual section-wise evaluations.
- The system should notify students via email about their grades.
- The system should produce result reports in PDF/Excel formats for export purposes.

Non-Functional Requirements

Performance

- The system must load exam pages within 2 seconds when 500 users access the system at the same time.
- Anti-cheating algorithms must detect and handle events in real-time with a delay of no more than 500 ms.

Scalability

- The system must support horizontal scalability to handle up to 10,000 concurrent users.

Availability

- The system must maintain operational status for 99.9% of the time during exam periods.

Security

- The system must use TLS 1.3 encryption protocols for transmitting information and AES-256 encryption for stored data.
- The system must protect user data from SQL injection, XSS, and CSRF vulnerabilities

Usability

- The student exam application must get a score of at least 80 in the SUS usability test.

Reliability

- The system shall auto-save student answers every 30 seconds to prevent data loss.

Maintainability

- The source code must follow the principles of SOLID design with at least 80 test coverage.

Compliance

- The system must comply with GDPR/PDPA regulations for storing biometric data.

Compatibility

- The system will operate properly on Chrome and Edge (all in use versions).

Accessibility

- The user interface will meet the accessibility guidelines of WCAG 2.1 level AA

Use Case Diagrams

The system has four primary actors and six major use case packages. The diagrams below describe the actor-system interactions in the OESCD.

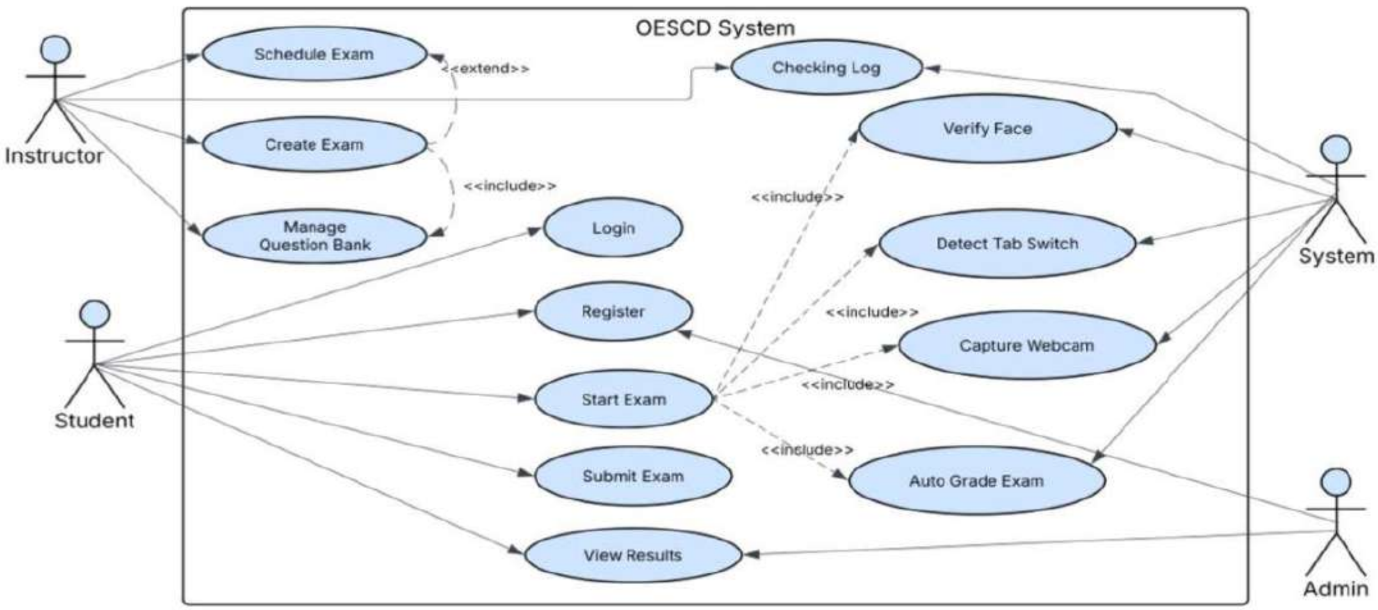

Figure 1 Use Case Diagram

Actors

- **Student:** Takes exams, views results, manages profile.
- **Instructor:** Creates exams, manages question banks, grades essays, reviews cheating logs.
- **Exam Administrator:** Manages exams, schedules exams, assigns students to exams.
- **System/AI Engine:** Supervises exam sessions, identifies cheating, generates statistics.

Use Case Packages

Table 1 Use Case Packeages

Package	Use Cases
Authentication	Login, Register, Reset Password, Logout, 2FA Verification
Exam Management	Create Exam, Edit Exam, Delete Exam, Schedule Exam, Assign Students
Question Bank	Add Question, Edit Question, Delete Question, Import Questions (CSV), Tag by Difficulty
Exam Taking	Start Exam, Answer Question, Flag Question, Submit Exam, Auto-Submit on Timeout
Cheating Detection	Capture Webcam, Verify Face, Detect Gaze Deviation, Detect Tab Switch, Log Violation, Generate Incident Report
Grading & Results	Auto-Grade MCQ, Manual Grade Essay, Release Results, Export Report, Send Notification

Use Case Description Tables

Student Login

Table 2 UC-01 Student Login

Use Case ID	UC-01
Use Case Name	Student Login
Actor(s)	Student
Description	A registered student enters credentials which enables them to access the exam portal.
Pre-conditions	The student possesses an active account. System is online.
Post-conditions	The system confirms the student's identity, which results in their automatic transfer to the dashboard.
Main Flow	i. The student accesses the login page. ii. The user submits their username and password information. iii. The system proceeds to check the user's credentials. iv. The system establishes a new session after successful credential validation and sends the user to their dashboard. v. The system dispatches one-time password information to the user's email address when two-factor authentication has been activated. vi. The student enters the one-time password to obtain access.
Alternate Flow	1. When users input incorrect login information the system shows an error message which permits them to attempt logging in five more times. 2. The system displays a lock message when accounts become locked and requires users to initiate password recovery.
Exceptions	The system presents a maintenance message because the server is currently down.
Business Rules	Password must be at least 8 characters. The system will block user access after they enter incorrect credentials five times.

Create Exam

Table 3 UC-02 Create Exam

Use Case ID	UC-02
Use Case Name	Create Exam
Actor(s)	Instructor
Description	The instructor establishes a new examination after completing all necessary setup work.
Pre-conditions	The authentication process confirms instructor identity existence. The question bank contains at least one question.
Post-conditions	The exam exists in either its draft form or its scheduled state.
Main Flow	i. Instructor clicks 'Create Exam'. ii. Fills in title, description, date/time window, duration.

	iii. Selects questions manually or via auto-select from bank. iv. The user establishes proctoring requirements through webcam control and tab-locking features. v. The user establishes a passing score requirement for the test. vi. The user establishes an examination draft which he/she can save or he/she can publish or schedule for actual testing.
Alternate Flow	1. The system uses auto-select to choose N questions which create a difficulty distribution of questions. 2. The exam remains as a draft version which students cannot see yet.
Exceptions	The system prevents publishing because no questions were added.
Business Rules	The exam duration requires a minimum length of 10 minutes and a maximum length of 300 minutes.

Take Exam

Table 4 UC-03 Take Exam

Use Case ID	**UC-03**
Use Case Name	Take Exam
Actor(s)	Student, System
Description	The student completes the assigned test while the system monitors his activities with real-time proctoring
Pre-conditions	The exam remains active because the student has already enrolled in the exam. The student has given permission to use both his webcam and microphone for the exam.
Post-conditions	The exam submission process completes when the student submits his answers which the system records and the proctoring log achieves finalization
Main Flow	i. The student accesses his exam through the dashboard. ii. The system conducts identity verification through face capture. iii. The browser system enters a complete full-screen lockdown mode. iv. The exam questions appear to students either as single items or as complete sets. v. The student has an available time frame to answer the questions. vi. The student submits his answers through the 'Submit' button or the system will automatically submit all answers when the time limit ends. vii. The system preserves answers while it concludes the session.
Alternate Flow	1. The student marks a question he/she wants to review before submitting his final examination. 2. The system identifies a lost connection which results in automatic answer preservation while students can reconnect during the designated time frame.
Exceptions	The face verification process must be completed before students can begin their exam.
Business Rules	The student needs permission from his instructor to take another attempt at the exam.

Detect Cheating

Table 5 UC-04 Detect Cheating

Use Case ID	**UC-04**
Use Case Name	Detect Cheating
Actor(s)	System (AI Engine)
Description	The system maintains continuous monitoring of test-takers throughout the exam while it records any activities that seem suspicious.
Pre-conditions	The examination process remains underway because proctoring monitoring has commenced for this particular test.
Post-conditions	The system records violation incidents with timestamps while it creates an incident report.
Main Flow	i. The System streams webcam feed through AI vision module. ii. AI checks for face presence and identity match every 5 seconds. iii. System monitors browser events for tab-switch minimize or copy-paste. iv. Gaze tracking module analyzes eye movement direction. v. If a violation is detected system logs event with screenshot / timestamp. vi. The student receives a warning overlay for minor violations. vii. The system terminates users who commit severe violations which include three or more face absences.
Alternate Flow	1. The system identifies an immediate severe violation when multiple facial features appear. 2. The system can only detect physical notetaking by students because it tracks screen and browser activity.
Exceptions	The system alerts the instructor about webcam hardware failure while it relies on browser-only monitoring.
Business Rules	The system terminates the exam when three warning events of the same type occur.

Grade Exam

Table 6 UC-05 Grade Exam

Use Case ID	**UC-05**
Use Case Name	Grade Exam
Actor(s)	System, Instructor
Description	The system automatically grades objective questions while human graders assess subjective questions.
Pre-conditions	The submission deadline for the exam has passed because one student has already submitted their work.
Post-conditions	The system calculates all grades which it stores until the final release of results.
Main Flow	i. System automatically grades multiple choice and true/false questions. ii. Instructor uses the manual grading tool to assess essay and short answer responses.

	iii. Instructor provides marks and comments for each question. iv. System determines final score together with passing and failing results. v. Instructor clicks 'Release Results'. vi. Students receive email notification and can view results on dashboard.
Alternate Flow	The instructor can use manual scoring to override automatic grading after content assessment.
Exceptions	The system proceeds to skip steps 2 and 3 because all questions can be graded automatically.
Business Rules	The instructor has the authority to award partial marks for short answers according to his personal judgment.

Review Cheating Report

Table 7 UC-06 Review Cheating Report

Use Case ID	**UC-06**
Use Case Name	Review Cheating Report
Actor(s)	Instructor, Exam Administrator
Description	The instructors examine the AI-created reports of cheating incidents which occurred during the exam.
Pre-conditions	The examination process has ended because at least one proctoring event was recorded.
Post-conditions	The instructor can choose between three actions: warning, invalidating the result, or clearing the student.
Main Flow	i. The instructor begins the process by opening the Proctoring Reports panel. ii. The system displays all students together with their recorded violation counts. iii. The instructor selects a student to examine the sequence of events which occurred. iv. The display shows each event through three elements: timestamp, type, and screenshot (which is optional). v. The instructor designates the report as 'Reviewed' status. vi. The instructor has the authority to either escalate the situation to the administrator or invalidate the student's test results.
Alternate Flow	The report shows a clean status because no violations were detected.
Exceptions	The event log only records browser-based violations because webcam data is missing.
Business Rules	A written explanation is required to accompany the process of result invalidation, which creates an unchangeable outcome.

Chapter 2

Context Diagram & Data Flow Diagram (DFD)

The Context Diagram displays the Online Exam System as a single process which connects to four outside entities:

- **Student** supplies login information, exam responses and webcam video. Students receive exam materials and examination results from the system.
- **Instructor** supplies exam setup information, testing material to the system, obtains grading results and academic dishonesty detection reports.
- **The Exam Administrator** supplies scheduling information to the system, obtains system performance data and current testing information.
- **The Email Service** system processes notification requests and dispatches result and OTP emails to users. [3]

The main process OESCD System controls all data exchanges between these entities and the internal data storage systems.

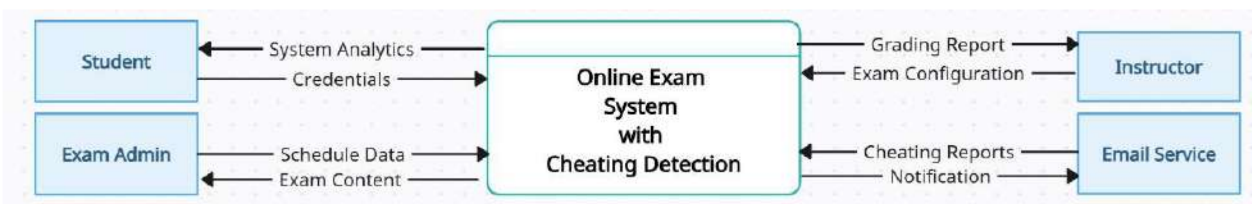

Figure 2 Context Diagram

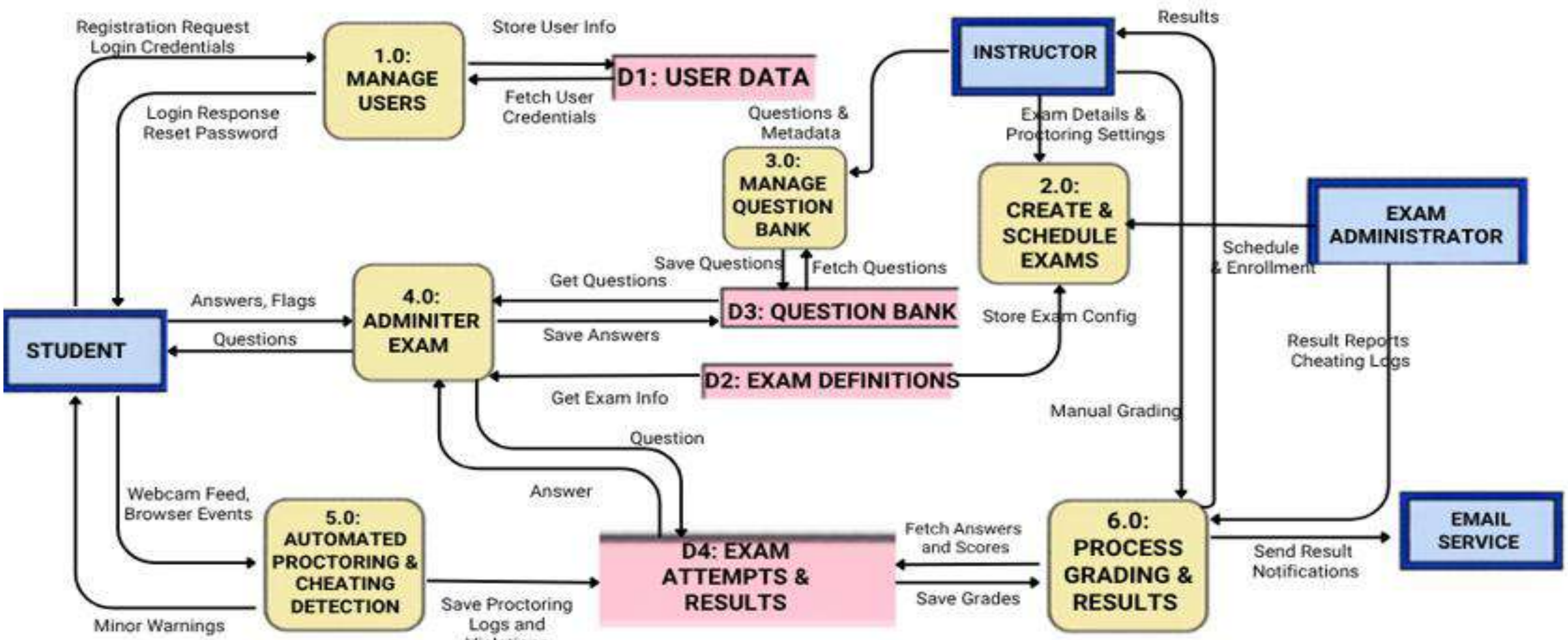

Figure 3 Level 0 Diagram

Activity Diagram

The class diagram for the OESCD system identifies key domain classes and their relationships

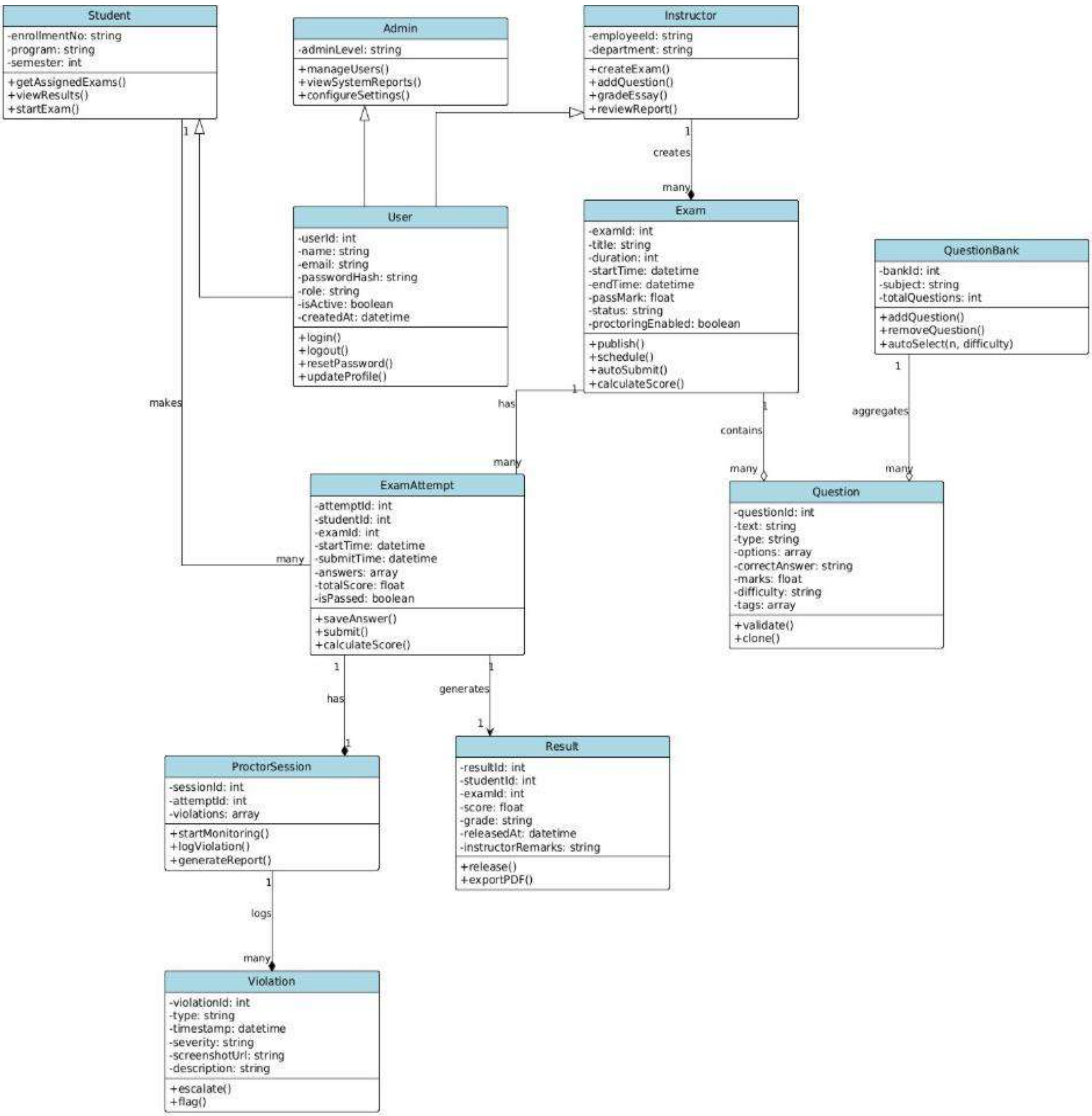

Figure 2 Class Diagram

Key Relationships

- **User → Student/Instructor/Admin:** Inheritance (Generalization).
- **Instructor → Exam:** Composition (one Instructor creates many Exams; Exam cannot exist without Instructor).
- **Exam → Question:** Aggregation (Exam includes Questions; Questions can exist independently in QuestionBank).
- **Student → ExamAttempt →** Exam: Association through ExamAttempt junction class.
- **ExamAttempt → ProctorSession:** Composition (one-to-one; ProctorSession is part of ExamAttempt lifecycle).
- **ProctorSession → Violation:** Composition (Violations belong to a ProctorSession).
- **ExamAttempt → Result:** Dependency (Result is derived from ExamAttempt data).

Sequence Diagram

The following sequence describes the interaction between objects when a student initiates and completes a proctored exam:

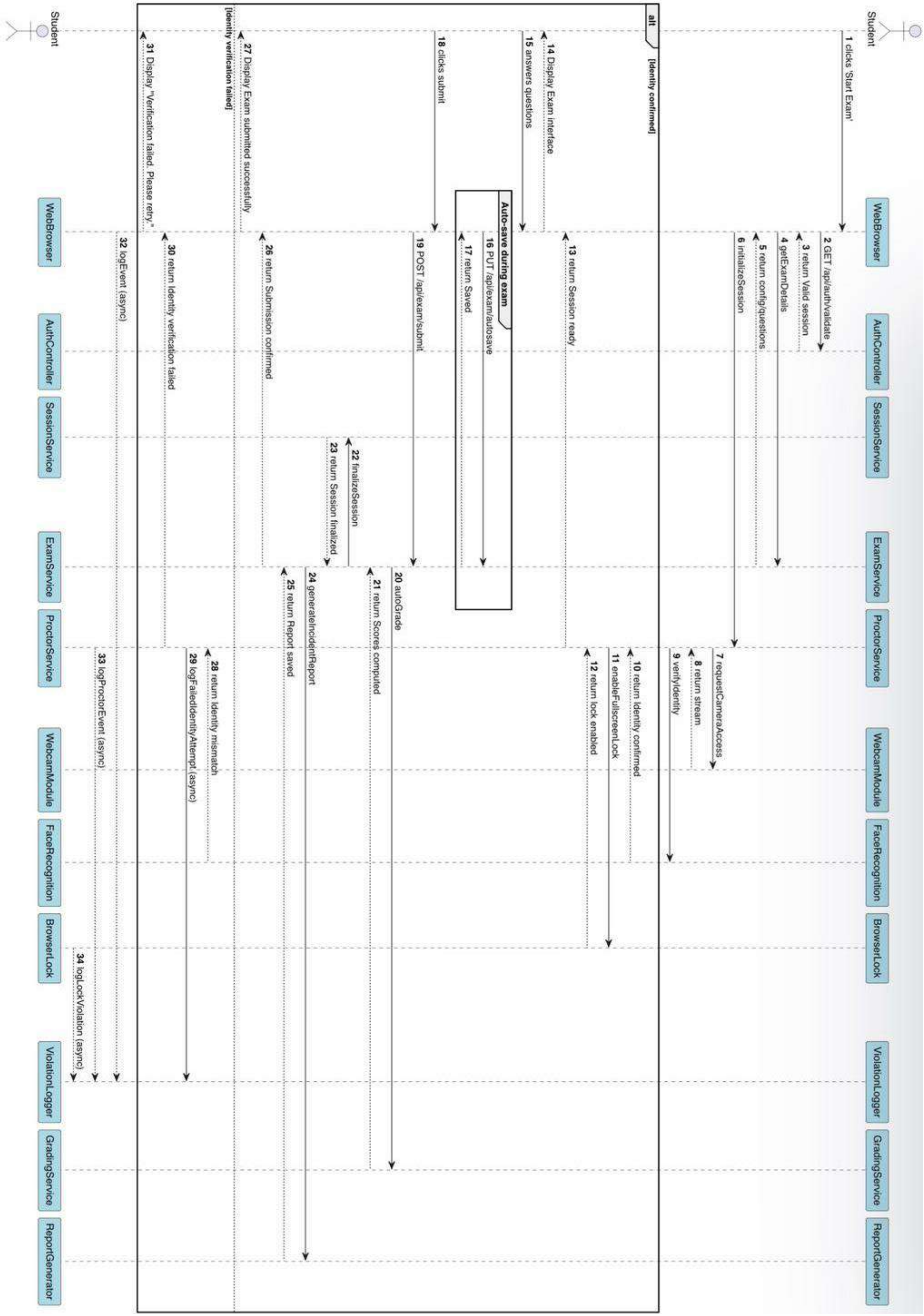

Figure 3 Sequence Diagram

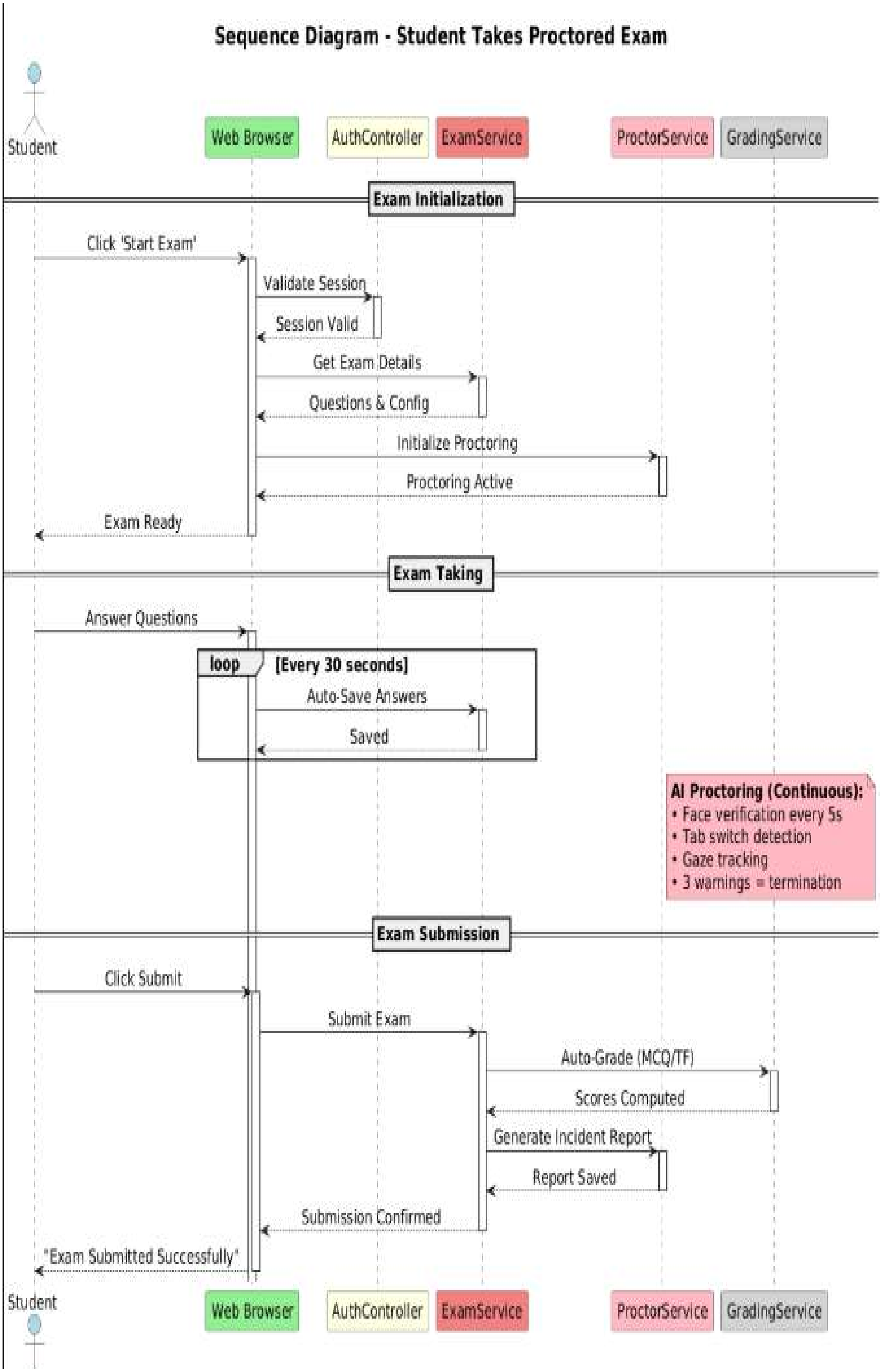

Figure 4 Sequence Digarm

Activity Diagram

The activity diagram traces the complete flow of a student's exam session from login to result notification. The workflow begins when a student logs in and chooses an exam. The exam window must remain open to permit access to the examination. The system begins proctoring when the window opens which uses AI technology to track facial presence, eye movements and tab switching activities. The system automatically grades multiple-choice questions after submission while instructors evaluate the shorter and essay questions. The student receives their final grade after publication or if told otherwise according to the instructions.

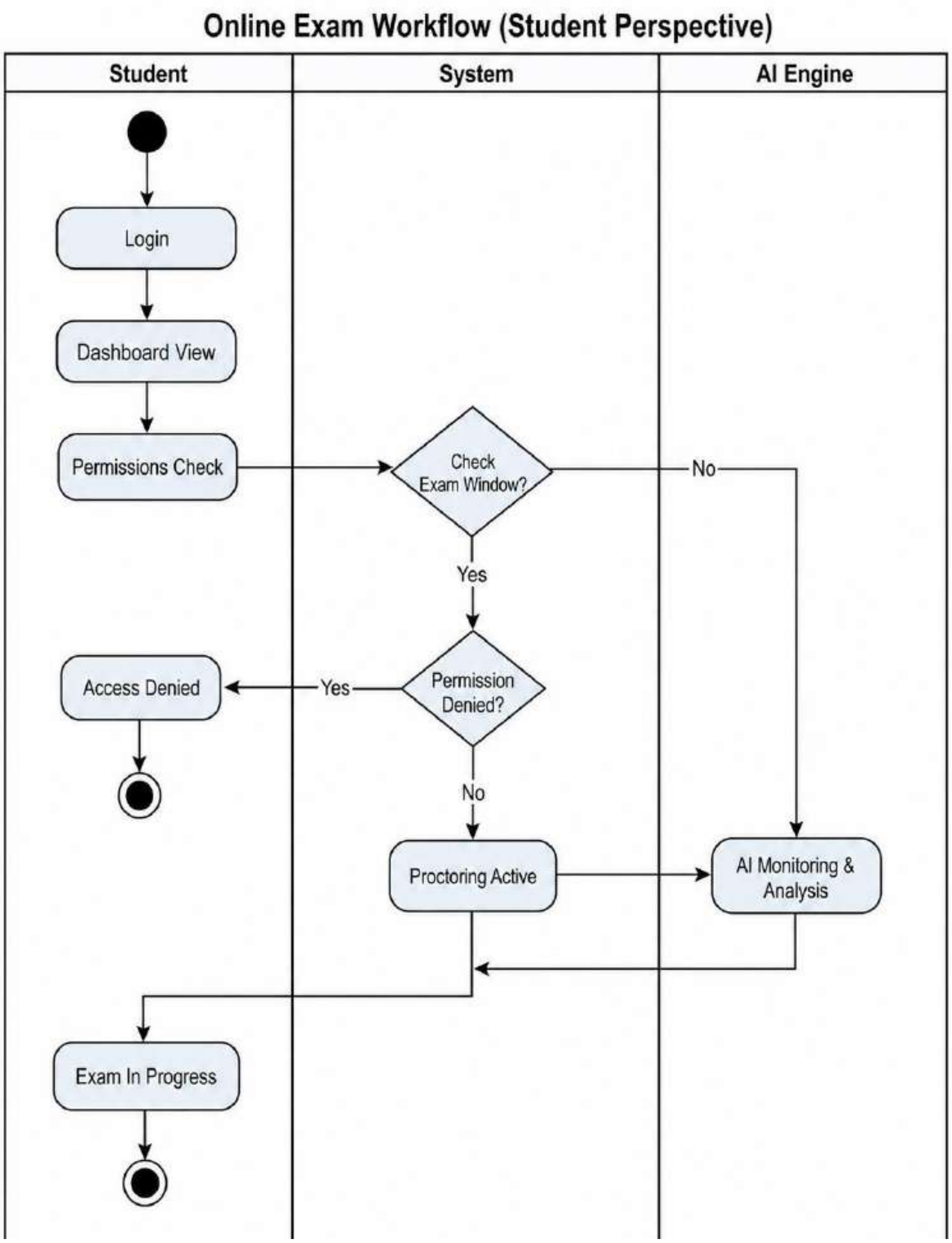

Figure 5 Activity Diagram

Chapter 3

Architecture diagrams:

MVC Architecture:

The frontend implements MVC through React Components which serve as the View to present the user interface while Event Handlers and Axios function as the Controller to handle user interactions and the Redux Store serves as the Model to control application state. The system transmits change notifications from the Model to the View whereas user events move from the View to the Controller and the Controller sends state updates to the Model. [4]

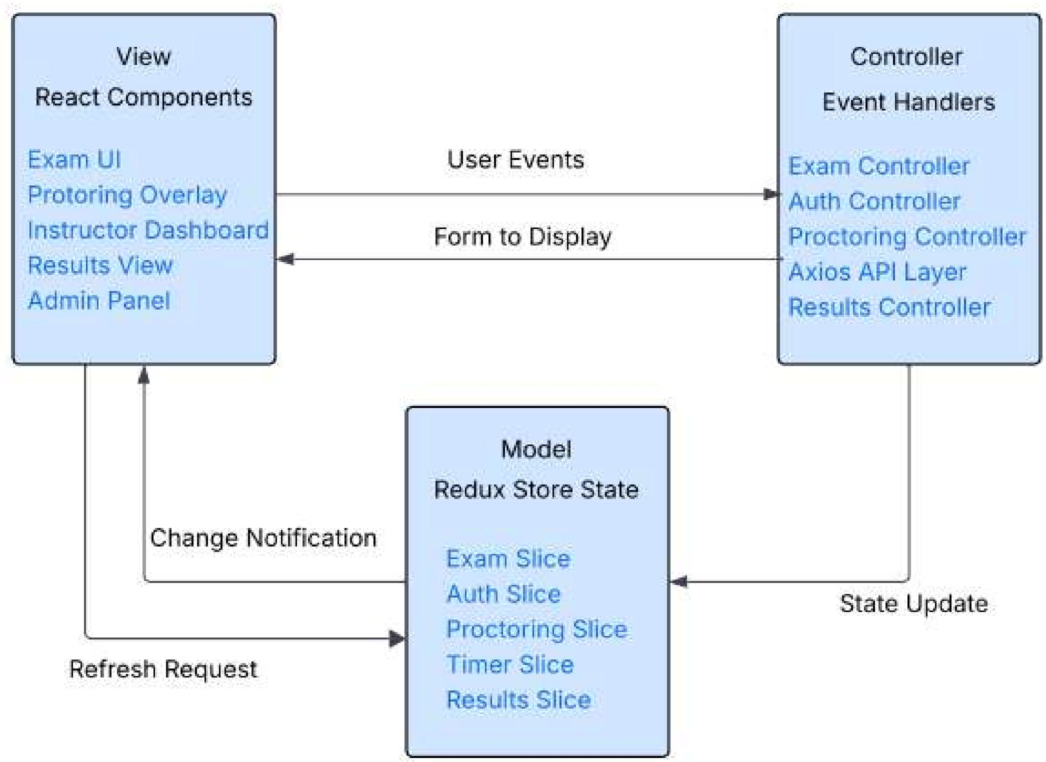

Figure 6 MVC Architecture Diagram

Layered Diagram:

The backend system consists of four horizontal layers which operate in different ways.

- The Presentation Layer handles API requests via controllers.
- The Business Logic Layer implements core functionality through services.
- The Data Access Layer abstracts database operations through its repository system.
- The Infrastructure Layer provides databases, cache, storage, and ML services.
- An API Gateway handles the request routing which connects the frontend and backend systems.

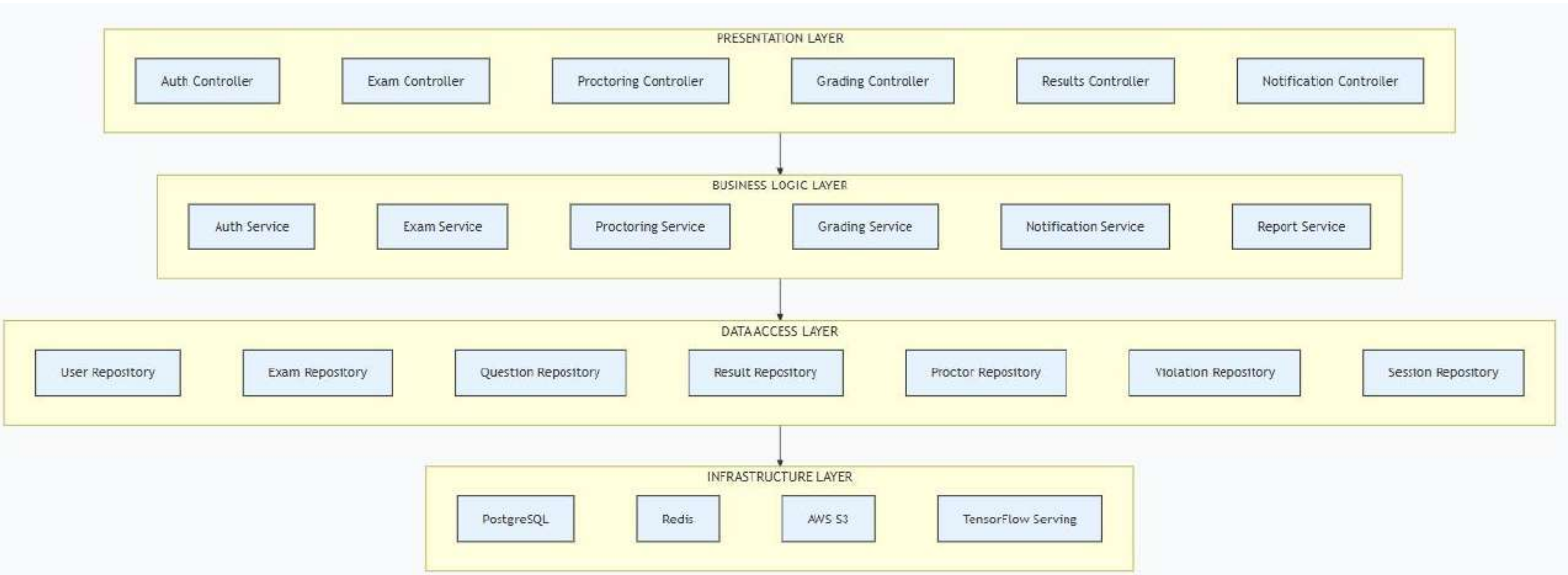

Figure 7 Layered Architecture Diagram

State Diagram

Exam Entity State Machine

The Exam Entity Lifecycle State Diagram illustrates the complete flow of an exam from creation to archival. The system specifies all possible states an exam can occupy together with the permitted transitions between those states which occur through system events and administrative actions and time-based triggers.

The Exam entity transitions through the following states during its lifecycle:

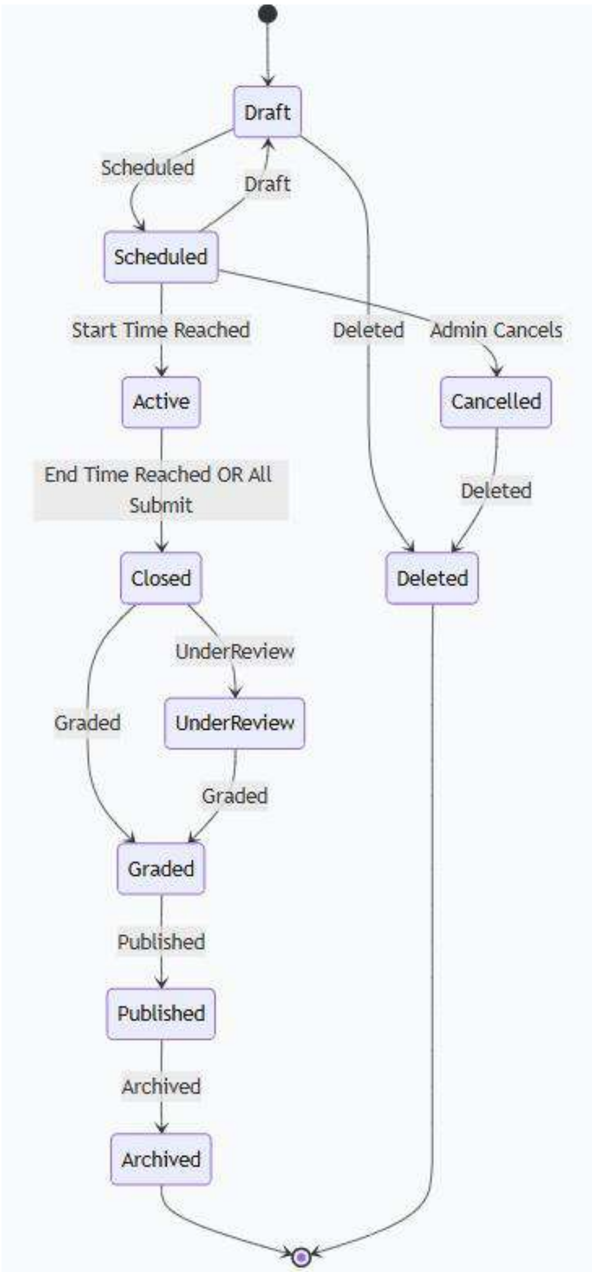

Figure 8 State Diagram

Chapter 4

1. Introduction

This test plan documents the testing strategy for the **TESTify Online Examination System with Cheating Detection (OESCD)**. The plan follows **IEEE** standard for software test documentation.

2. Test Objectives

Table 8 Test Objectives

Type	Objective
Validation Testing	Demonstrate that TESTify meets all functional and non-functional requirements
Defect Testing	Discover faults in cheating detection, exam workflow, and grading mechanisms

3. Test Levels

Table 9 Test Levels

Level	Scope	Responsibility
Unit Testing	Individual methods (e.g., verifyFace(), detectTabSwitch())	Developers
Integration Testing	Interactions between components (e.g., ProctorSession → Violation logging)	Development Team
System Testing	End-to-end scenarios (e.g., complete exam lifecycle)	Testing Team

4. Unit Test Cases

Tests targeting individual methods and functions in isolation.

Unit Test 1: Face Verification Module

Table 10 UT-01 Face Verification Module

Test Case ID	UT-01
Description	Verify that verifyFace() correctly matches student face with stored profile
Pre-conditions	Webcam is functional; student profile has enrolled face data
Test Data	Positive: Correct student face Negative: Different person's face Invalid: No face detected
Test Steps	1. Call verifyFace(captureFrame) 2. Compare against enrolled template 3. Return confidence score
Expected Result (Positive)	Confidence > 0.85 ; returns TRUE
Expected Result (Negative)	Different face; returns FALSE with "Face mismatch" error
Expected Result (Error)	No face; returns FALSE with "No face detected"
Actual Result	
Status (Pass/Fail)	

Unit Test 2: Tab Switch Detection

Table 11 UT-02 Tab Switch Detection

Test Case ID	UT-02
Description	Verify that detectTabSwitch() logs violation when user leaves exam window
Pre-conditions	Browser lockdown enabled; Exam session active
Test Data	Positive: User switches to another tab Negative: User stays on exam tab
Test Steps	1. Simulate visibilitychange event 2. Call detectTabSwitch() 3. Check violation count
Expected Result (Positive)	Violation logged ; Violation count +1, warning overlay shown
Expected Result (Negative)	No violation logged
Expected Result (Boundary)	After 3 violations : exam auto-submitted
Actual Result	
Status (Pass/Fail)	

5. Integration Test Cases

Tests verifying correct interactions between system components.

Integration Test 1: ProctorSession: Violation Logging

Table 12 IT-01 ProctorSession:Violation Logging

Test Case ID	IT-01
Description	Verify that cheating events detected by ProctorSession are correctly saved to the Violation table
Pre-conditions	Database connected; Exam attempt in progress
Test Data	Simulated tab-switch event; face absence event
Test Steps	1. ProctorSession detects violation 2. Calls logViolation(violationType, timestamp, screenshot) 3. Query database for violation record
Expected Result	Violation record created with correct foreign key (ProctorSession ID); timestamp within 500 ms of detection
Actual Result	
Status (Pass/Fail)	

Integration Test 2: Exam Submission: Grading Pipeline

Table 13 IT-02 Exam Submission: Grading Pipeline

Test Case ID	IT-02
Description	Verify that submitted exam answers flow correctly to auto-grader and manual grading queue
Pre-conditions	Exam contains MCQs and essay questions; Student has submitted
Test Data	10 MCQs (8 correct), 2 essay answers
Test Steps	1. Student clicks Submit 2. System calls autoGrade(answers) for MCQs 3. System creates manual grading task for essays 4. Instructor grades essays

Expected Result	MCQ score computed (8/10 = 80%); Essays appear in instructor's grading queue; Final grade = MCQ score + essay score
Actual Result	
Status (Pass/Fail)	

6. System Test Cases

End-to-end tests covering complete use-case scenarios.

System Test 1: UC-01 Student Login (Positive & Negative)

Table 14 ST-01 Student Login (Positive & Negative)

Test Case ID	ST-01
Use Case Reference	UC-01: Student Login
Test Type	Validation & Defect
Description	Verify login functionality with correct and incorrect credentials

Scenario	Input Data	Expected Result
Positive: Valid credentials	Email: student@university.edu Password: Pass@1234	Redirect to dashboard; Session created
Negative: Wrong password	Same email Password: WrongPass	Error: "Invalid credentials"; No session created; Counter increments
Negative: Locked account	After 5 failed attempts + 6th attempt	Error: "Account locked. Reset password"
Negative: Empty fields	No email, no password	Error: "Email and password required"
Boundary: Password length	Password = 7 characters	Error: "Password must be at least 8 characters"
Exception: Server down	Any credentials	Error: "Service unavailable. Try later"

System Test 2: UC-03 Take Exam (End-to-End)

Table 15 ST-02 Take Exam (End-to-End)

Test Case ID	ST-02
Use Case Reference	UC-03: Take Exam
Test Type	Validation & Defect
Description	Verify complete exam-taking experience with proctoring

Scenario	Test Steps	Expected Result
Positive: Normal exam	1. Student logs in 2. Selects "Midterm Exam" 3. Face verification passes 4. Answers 20 questions 5. Clicks Submit	All answers saved; Submission confirmed; Grade processing initiated
Positive: Auto-submit on timeout	1. Start 10-minute exam 2. Do not submit 3. Wait for timer to reach 0	System auto-submits answers; Student notified
Negative: Face verification fails	1. Start exam 2. Webcam shows different face	Exam does not start; Error: "Face verification failed"
Negative: Network disconnect	1. During exam, disconnect WiFi 2. Reconnect within 2 minutes	Answers auto-saved locally; Resume without data loss
Boundary: Flagged question	Student clicks "Flag" on Q5	Question marked for review; Navigation works

System Test 3: UC-04 Detect Cheating (Critical)

Table 16 ST-03 Detect Cheating (Critical)

Test Case ID	ST-03
Use Case Reference	UC-04: Detect Cheating
Test Type	Defect Testing
Description	Verify AI proctoring detects all cheating behaviors as per requirements

Scenario	Test Steps	Expected Result
Positive: Tab switching	During exam, press Alt+Tab to switch browser tab	Violation logged; Warning overlay: "Tab switching detected"
Positive: Face absence	Student leaves webcam view for 10 seconds	Violation logged; Warning: "Face not detected"
Positive: Multiple faces	Another person enters webcam frame	Severe violation logged; Exam terminated immediately
Positive: Eye gaze deviation	Student looks away from screen repeatedly	Gaze tracking logs violation; Warning issued
Negative: Normal behavior	Student maintains eye contact, no tab switch	No violations logged; Exam continues
Boundary: Three warnings	Trigger tab-switch 3 times	Exam auto-submitted; Student locked out
Exception: Webcam failure	Webcam hardware stops working	System alerts instructor; Falls back to browser-only monitoring

System Test 4: UC-05   Grade Exam

Table 17 ST-04 Grade Exam

Test Case ID	ST-04
Use Case Reference	UC-05: Grade Exam
Test Type	Validation & Defect
Description	Verify auto-grading and manual grading produce correct results

Scenario	Test Steps	Expected Result
Positive: Auto-grade MCQs	Student answers: 15/20 correct	Score = 75%; Displayed instantly
Positive: Manual grade essay	Instructor assigns 7/10 to essay	Essay score = 7; Final score updated
Positive: Release results	Instructor clicks "Release Results"	Student receives email; Results visible on dashboard
Positive: Export report	Instructor clicks "Export as PDF"	PDF generated with student name, scores, section breakdown
Negative: Override auto-grade	Instructor changes MCQ score from 8 to 9	Override saved; Audit log records change
Boundary: Partial marking	Short answer partially correct	Instructor assigns 3/5; System accepts partial marks
Exception: No essay grader	No instructor assigned for essay questions	System queues essays; Notifies admin to assign grader

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